

## ITS122P Group 1 Project Proposal | Printokids PH

### I. Introduction

#### Client Background

Printokids PH is a small business based in the Philippines that focuses on personalized printing services and customized souvenirs for different occasions and celebrations. The business provides creative, affordable, and high-quality products that help customers make their events more meaningful and memorable. From customized merchandise to event giveaways and souvenirs, Printokids PH offers different products and services that match the style and preferences of their clients. Printokids PH ensures every project piece reflects the unique style, theme, and name of the recipient or occasion whether it's a wedding, birthday, corporate events, or simple celebration event which needs to have quality souvenirs, or gifts.

Behind Printokids PH is a 31-year-old businesswoman who has a strong passion for crafting and creativity. Her love for designing and making personalized items inspired her to start the business in late 2024. As a mother, she also enjoys spending time with her children through arts and crafts activities, especially decorating different items together. This family bonding experience became one of the inspirations behind Printokids PH, encouraging her to turn her creativity into a business that spreads joy and memorable experiences to others. Through her dedication, artistic skills, and passion for personalized creations, she continues to grow the business while providing meaningful and high-quality products for every customer and occasion.

#### Products / Services Offered

The business specializes in printing on items such as tote bags, mugs, stickers, labels, tags, and other personalized products. They also create souvenirs like keychains, customized stickers and magnets, and giveaways for birthdays, weddings, corporate events, and other special gatherings. One of the unique services offered by Printokids PH is their PICK AND PRESS interactive event booth where guests can personally decorate tote bags, small purses, and other items using stickers and creative materials. This on-the-spot personalization activity gives guests a fun and memorable experience while allowing them to create their own customized souvenirs.

They blend artistic skills with high-quality printing and craftsmanship, offering versatile solutions that suit both small and big events. Whether you're looking to add a personal touch to giveaways, creating one of a kind souvenirs, or design branded merchandise, Printokids PH also promotes ON-THE-SPOT SOUVENIRS, and their team brings ideas to life with precision and care. Affordable yet premium quality, their services make personalized gifting accessible, helping clients express gratitude, celebrate milestones, and leave a lasting impression on their guests and loved ones.

What makes Printokids PH different from other small printing businesses is the warm and personal experience they give to their customers. The business does not only focus on creating customized products, but also values the memories and emotions connected to every item they make. The team believes that souvenirs are more than just simple giveaways because they serve as keepsakes that help people remember special moments and celebrations. Through personalized items such as mugs, tote bags, keychains, stickers, and event giveaways, Printokids PH aims to give clients and their guests something meaningful that they can keep and treasure for a long time. With their

passion for creativity, crafting, and personalization, the business continues to create products that make every celebration more memorable and heartfelt.

## **II. Problem Statement and Objectives**

As Micro, Small, and Medium Enterprises (MSMEs) scale, they frequently face operational bottlenecks where a minimal team or a single business owner must manually execute every core process from inventory tracking to order fulfillment to sustain operations. Printokids PH, is a specialized customized printing and gifting business that heavily experiences this manual workload. Currently, the business relies on its social media pages as its primary storefront, leading to an unautomated, fragmented workflow.

Customers seeking personalized merchandise, custom souvenirs, or interactive event services must navigate unstructured channels to search for products, place orders, and inquire about order statuses. This lack of automation forces the business owner to manually cross-reference order details against available raw materials and stock levels, increasing the risk of human error, delayed tracking updates, and limited customer reach. To aid in decluttering and simplifying these operations, an integrated, dynamic web application with a centralized database system is required to transition Printokids PH into an automated B2C model.

The following objectives shall be attained upon the website's completion:

- **Create a secure web system** for both customer-facing transactions and administrative operations.
- **Allow all website users** to browse customized product categories, view interactive booth services, and seamlessly search available items.
- **Allow registered clients** to securely log into their accounts, initiate customized orders, and input specific preferences.
- **Allow website admins** to execute full CRUD functionalities over the product catalog and centralized stock inventory.
- **Allow website admins** to manage user access handles and access consolidated sales and customer purchase histories.

## **III. UN SDG's Addressed**

A dynamic web application development for Printokids PH fulfills a number of United Nations Sustainable Development Goals (SDGs) that corresponds to small business development, responsible production, and community impact:

Goal 4: Quality Education (Indirect) - Creativity and the development of skills among event participants are fostered by interactive on-the-spot souvenir creation.

Goal 8: Decent Work and Economic Growth - The platform enhances operational efficiency for the business and promotes sustainable development by automating order and inventory management.

Goal 9: Industry, Innovation, and Infrastructure - Introduce technological innovation by implementing a centralized web system and database, which enables the business to expand while enhancing service delivery.

Goal 10: Reduced Inequalities - The system facilitates equitable engagement with Printokids PH services by offering customers the ability to customize their events and place orders over the internet.

Goal 12: Responsible Consumption and Production - Waste is minimized through the implementation of efficient stock management and order monitoring, which guarantees the responsible utilization of production resources and materials.

Overall, the suggested system supports sustainable growth, innovation, and equal access, while also encouraging responsible production and skill-building among users, illustrating how small-business digital transformation can advance multiple UN SDGs.

#### IV. Scope of the Project

For Printokids PH, this web application offers a centralized platform that enables clients to browse personalized product categories, place and track orders, and interact with interactive event booth services. Administrators manage inventory, approve orders, and maintain client accounts through a secure, responsive system with a centralized database, thereby streamlining operations and improving efficiency. This is achieved within the constraints of manual payment processing and physical event management, which are managed outside the platform.

#### V. Functional / Non-Functional Requirements

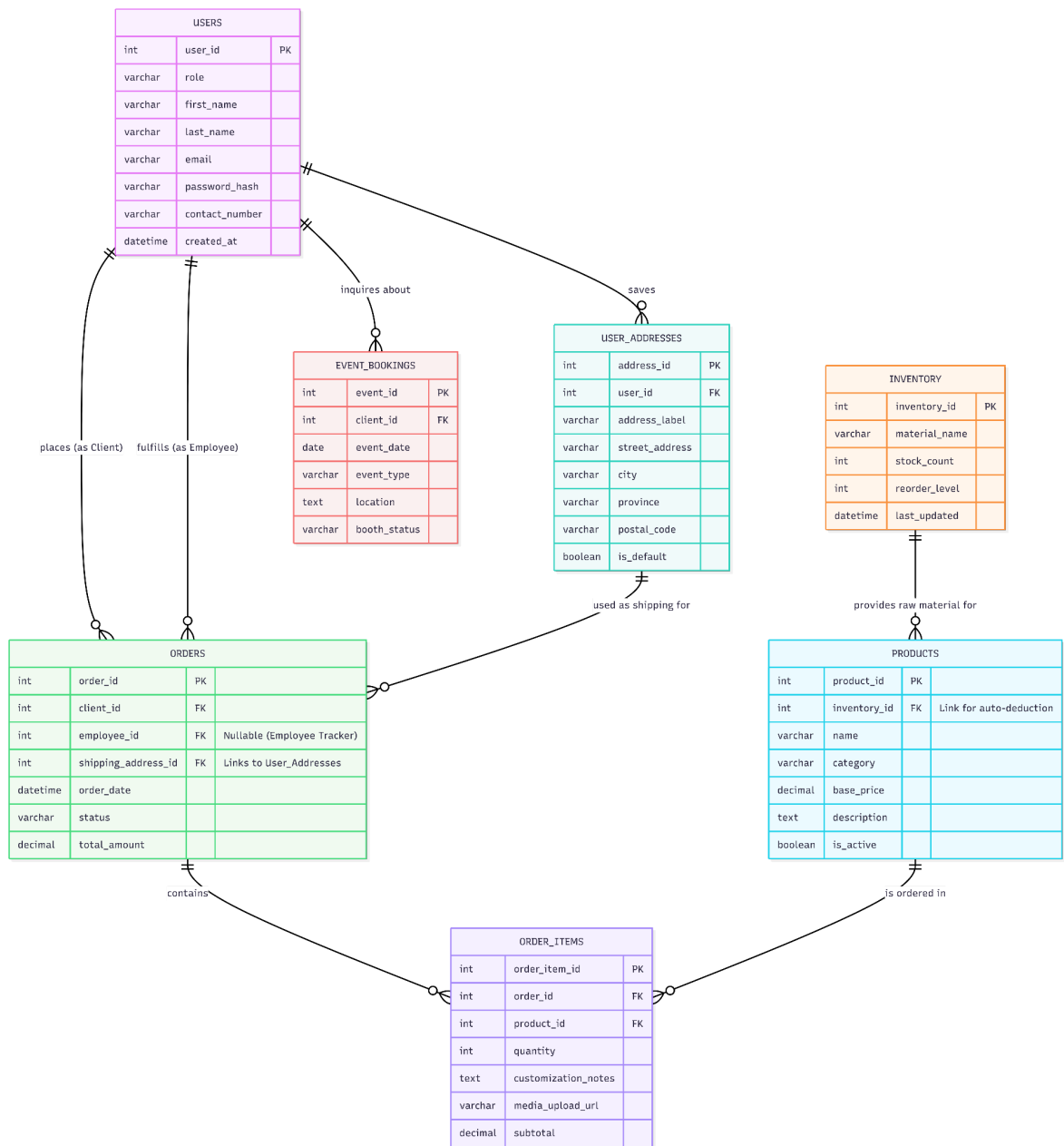
| Name                                      | Requirements                                                                                                             |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Functional Requirements</b>            |                                                                                                                          |
| <b>User Authentication</b>                | The website must allow users to securely register, log in, and reset their passwords when needed.                        |
| <b>User Profile Management</b>            | Registered users must be able to view, manage, and update their account profiles.                                        |
| <b>Product Display and Categorization</b> | The website must display a catalog of products categorized by product types.                                             |
| <b>Search and Filter</b>                  | The system must allow users to seamlessly search for specific items and filter products by category or theme.            |
| <b>Order Customization Input</b>          | The website must provide input fields for customers to submit any specific personalization or customization preferences. |

|                                          |                                                                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Custom Media Upload</b>               | The website must allow customers to securely upload design assets, images, or logos for personalized items during the ordering process.          |
| <b>Digital Shopping Cart</b>             | The system must offer a shopping cart interface allowing users to add, remove, and adjust quantities of customized items before checking out.    |
| <b>Automated Tracking System (ATS)</b>   | The system must provide a customer dashboard to monitor order lifecycles and shipment statuses in real-time.                                     |
| <b>Product Catalog Management (CRUD)</b> | The system must allow administrators to Create, Read, Update, and Delete (CRUD) product listings, pricing, and descriptions.                     |
| <b>Inventory Stock Control (CRUD)</b>    | Website admins must be able to execute full CRUD operations over raw materials and track stock counts in a centralized database.                 |
| <b>Inventory Auto-Deduction</b>          | The system must automatically deduct any corresponding raw materials from the centralized inventory database upon successful order finalization. |
| <b>Order Fulfillment Control</b>         | Admins must be able to manage order states (e.g., Pending, In Production, Shipped, Completed) and bind shipment tracking numbers.                |

| Name                                 | Requirements                                                                                                                                                                                                                      |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Non - Functional Requirements</b> |                                                                                                                                                                                                                                   |
| <b>Product Requirements</b>          |                                                                                                                                                                                                                                   |
| <b>Performance</b>                   | The website must load quickly. Database queries for inventory and transaction records must execute within 2 seconds (under normal load).                                                                                          |
| <b>Usability</b>                     | The website must be intuitive for both administrators and clients by applying clear GUI design principles. Users should be able to understand and navigate the system effectively upon first interaction, without prior training. |

|                                   |                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Compatibility</b>              | The website must operate reliably and remain fully compatible with all major modern web browsers, including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.                                                                                                                                   |
| <b>Dependability Requirements</b> |                                                                                                                                                                                                                                                                                                           |
| <b>System Robustness</b>          | The system must recover within 5 minutes after a crash.                                                                                                                                                                                                                                                   |
| <b>Data Integrity</b>             | <p>There must be no duplicate or orphaned records allowed between inventory and transaction data.</p> <p>All data deletion and modification operations shall undergo validation and authorization checks before being committed to ensure data integrity and compliance with access control policies.</p> |
| <b>Security Requirements</b>      |                                                                                                                                                                                                                                                                                                           |
| <b>Confidentiality</b>            | Passwords must not be freely visible. Only authorized personnel shall be able to access these.                                                                                                                                                                                                            |
| <b>Authentication</b>             | Clients and business personnel must input correct login credentials to access services. Services shall deny access to any accounts that have no validation.                                                                                                                                               |
| <b>External Requirements</b>      |                                                                                                                                                                                                                                                                                                           |
| <b>Legislative Requirements</b>   | <p>The website must comply to protect users' personal data as mandated by the Data Privacy Act of 2012 (RA10173).</p> <p>The website shall ensure accurate and non-deceptive representation of inventory and transaction data details in compliance with the Consumer Act (RA 7394).</p>                  |

## VI. Database Design (ERD)



To effectively transition PrintoKids PH from a manual, fragmented social media storefront to an automated Business-to-Consumer (B2C) model, a centralized relational database has been designed. The Entity Relationship Diagram (ERD) model illustrates the 7-table schema, structured to ensure data integrity, eliminate operational bottlenecks, and support the business's specialized and customized printing services.

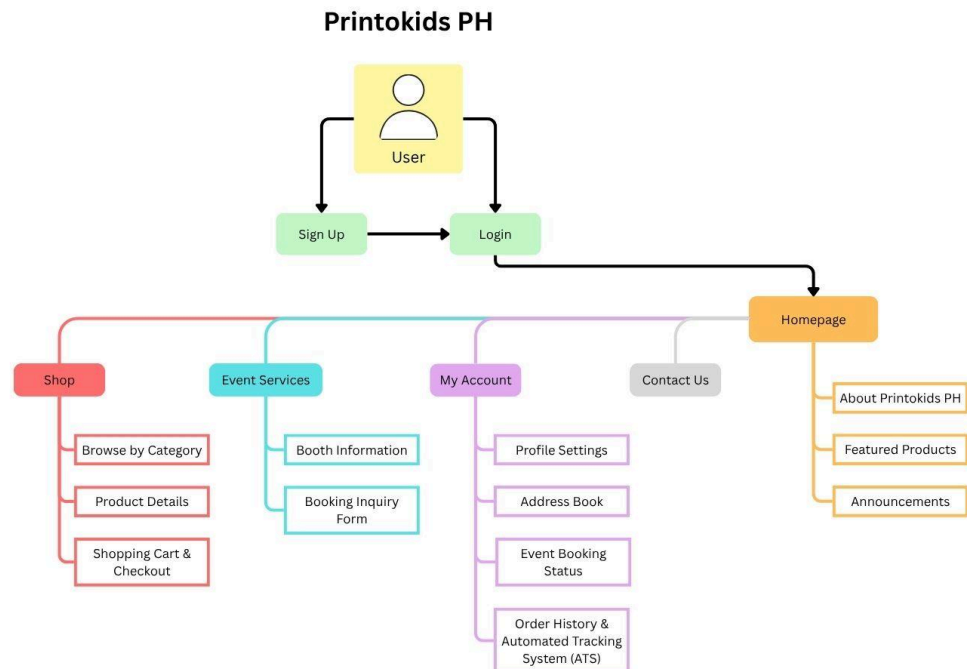
- **Users & User\_Addresses:** The **USERS** table serves as the secure, central hub for authentication and profile management, distinguishing between clients, employees, and administrators. To adhere to database normalization standards, shipping details have been moved to a dedicated **USER\_ADDRESSES** table. This one-to-many relationship allows a single user to securely save multiple delivery locations (e.g., Home, Office), thereby improving the checkout process.

- **Product & Inventory:** The INVENTORY table tracks the exact stock counts of raw materials (e.g., canvas tote bags, mugs). It shares a one-to-many relationship with the PRODUCTS table, which houses the active catalog products and their required raw materials via a foreign key, so the system can automatically deduct stock when an order is finalized, resolving the business owner’s dilemma of manually cross-referencing available stock. This process also supports sustainable material management.
- **Order & Order\_Items:** The transaction lifecycle is managed by the ORDERS table, which records the total amount, status, and selected shipping address. To track the operational fulfillment, orders are linked to both the placing client and the assigned employee. The ORDER\_ITEMS junction table connects these orders to the product catalog. Crucially, this table includes a customization\_notes field and a media upload URL, ensuring that Printokids PH customers' highly personalized preferences are securely captured for each item.
- **Event\_Bookings:** A dedicated table manages the inquiries for the ‘Pick and Press’ interactive event booth. It links directly to the USERS table, allowing clients to schedule physical event dates, specify locations, and track their booking approval.

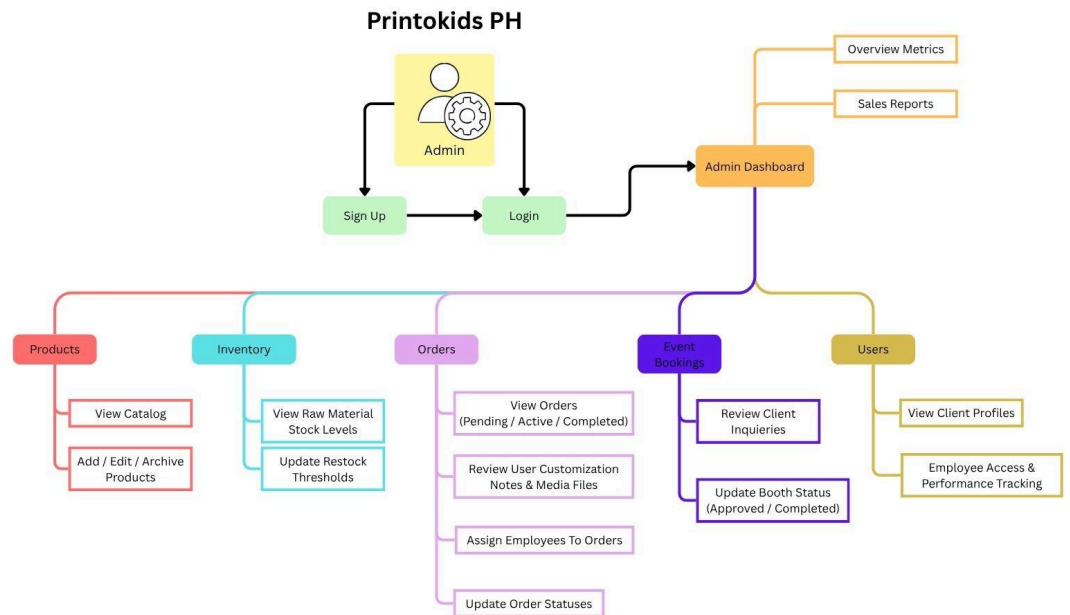
## VII. Features of the System

### A. Sitemap

#### I. Client-Facing Portal (Public & Registered Users)



## II. Administrative Backend (Admins & Employees)



### B. Sitemap Description:

The sitemap serves as the structural blueprint for the Printokids PH web application, dividing the platform into a seamless customer-facing portal and a secure administrative backend. The navigation flow is directly informed by the underlying relational database to ensure all user inputs are accurately captured and managed.

#### Client-Facing Navigation

- The user journey begins on the Home Page, which establishes the brand's creative identity and highlights featured services. Users navigate to the Shop to browse categorized merchandise. Upon selecting a product, the interface dynamically prompts the user to input their specific customization\_notes and upload optional design assets before adding the item to their cart.
- During checkout, users are seamlessly directed to select or input their delivery details, which are then saved to their profile in the USER\_ADDRESSES database.
- The Event Services page provides a dedicated form for clients to submit booking inquiries for the "PICK AND PRESS" booth. Finally, registered clients have access to My Account, a personalized dashboard where they can manage their saved addresses and utilize the Automated Tracking System (ATS) to monitor the real-time status of their customized orders and event bookings.

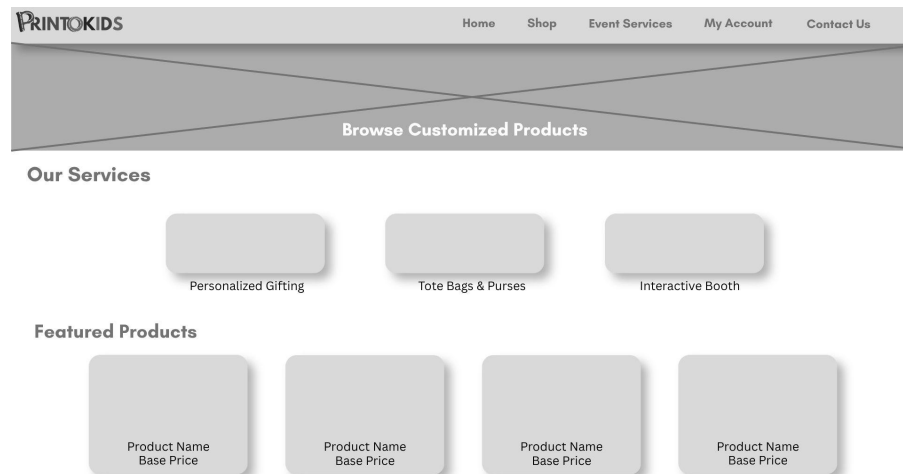


## Administrative Navigation

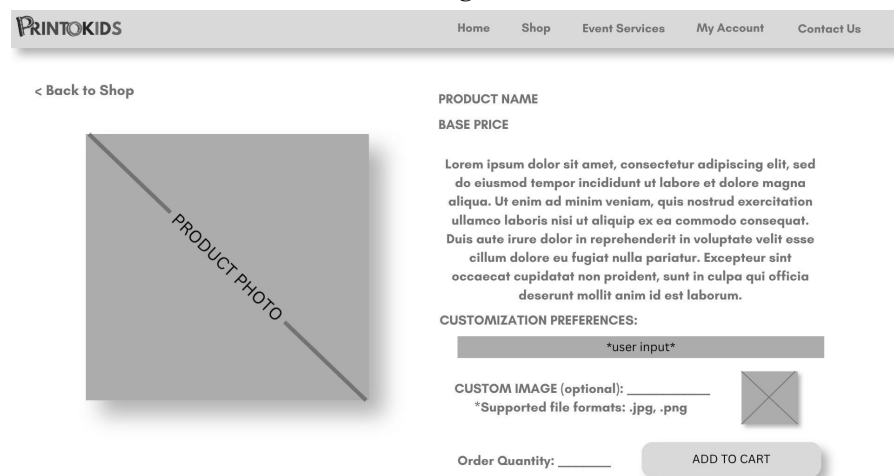
- Authenticated personnel are routed to a secure Admin Dashboard that serves as the control center for all business operations.
- The Manage Products and Manage Inventory sections provide administrators with full CRUD (Create, Read, Update, Delete) capabilities over the catalog and raw materials. Because the database links these two elements, admins can rely on the system to auto-deduct stock and trigger low-stock alerts.
- The Manage Orders module is the operational core, allowing the business owner to view captured customization notes, assign specific employees to fulfillment tasks, and update order statuses.
- Additionally, the Manage Users and Manage Events tabs allow the administration to oversee client accounts, evaluate employee performance histories, and approve interactive booth schedules.

## C. Wireframes

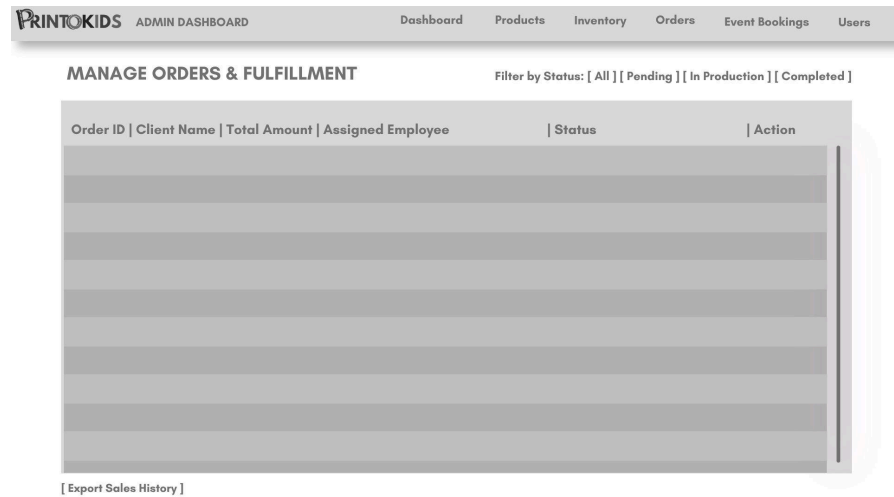
### 1. Home Page (Client View)



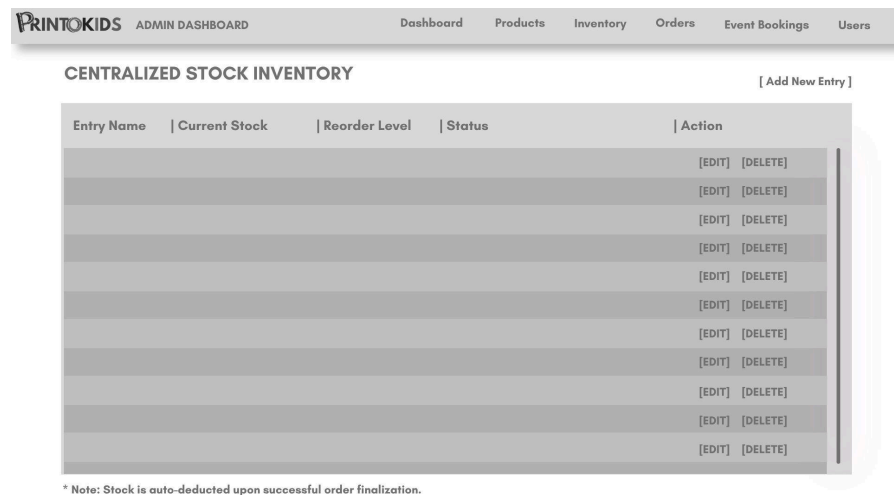
### 2. Product Details and Customization Page



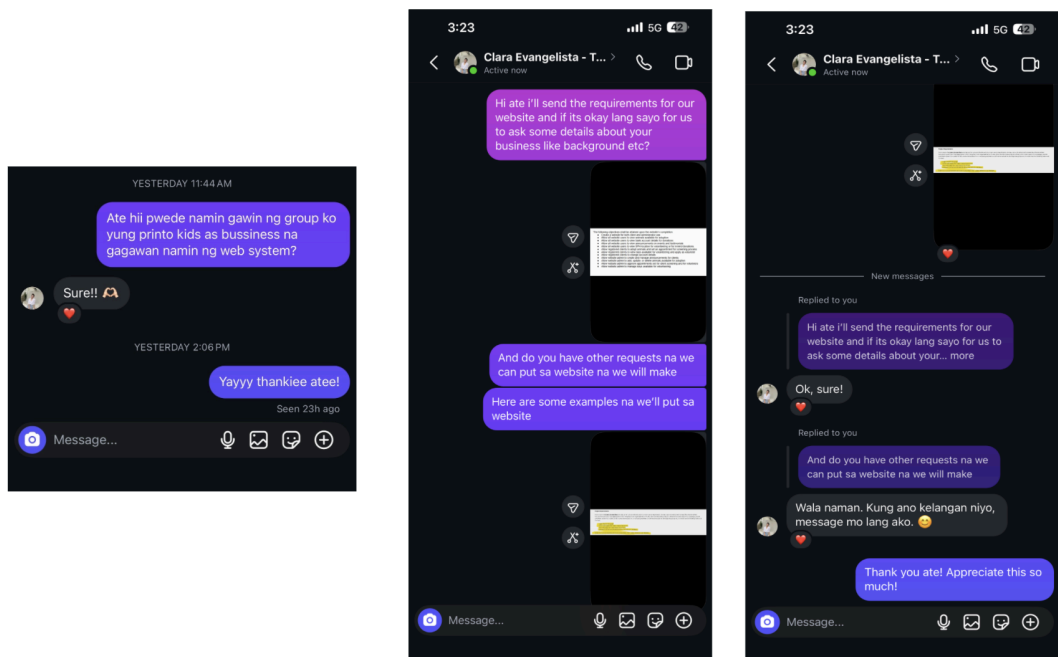
### 3. Admin Dashboard: Order Management and Admin View



### 4. Admin Inventory Control Dashboard



## VIII. Communication Letter



Initial communication was established by group member, Fiona Mykee R. Enriquez, who personally knows the business owner. A Facebook Messenger inquiry was sent to inquire about the project and deals.

## **IX. Acceptance Letter**

Info from additional discussions from the client and personal knowledge from Fiona helped inform the objectives discussed in the acceptance letter. The letter was written by the group, and then sent to the client to be signed for approval.

### **ACCEPTANCE LETTER**

To Whom It May Concern,

I, Clara Evangelista-Tolentino, representing PrintToKids PH, hereby confirm my acceptance of the proposed project to create a website and database for my business. Submitted by student group 1 for ITS122P from Mapua University.

I agree to collaborate with the student team composed of: Franco Emmanuel T. Castillo, Fiona Mykee R. Enriquez, Jose II R. Del Carmen, Josel Henri L. Caballero, Joshua M. Naive, and Vernadith C. Senin.

I acknowledge the objectives discussed:

- Create a website which will help me track key business processes (inventory management, transaction management, employee tracking).
- Create a database which will integrate my product inventory and transaction history.

CLIENT'S SIGNATURE:

A handwritten signature in black ink on a yellowed, lined paper background. The signature is stylized and appears to read 'Clara' followed by a long horizontal stroke and a dot.

PRINT NAME: Clara Evangelista-Tolentino

DATE SIGNED: May 29, 2026